Two transit candidates on KIC 6699887

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ABSTRACT

This note reports two transit candidates on KIC 6699887, a star Kepler recorded only in other targets' pixels: a 4.02 hour, 1,160 ppm dip every 15.113696 days over 81 transits and a 5.47 hour, 993 ppm dip every 47.69057 days over 28. In difference images both dips are nearer this star than its brightest neighbour in 4 of 4 observing seasons. The planets would be about 2.9 and 2.3 Earth radii. Null searches give false-alarm rates of 1 in 290 and 1 in 120 per star; TRICERATOPS 1.1.0 gives false-positive probabilities of 0.028 and 0.037 (no validation class applies at 30-minute cadence; Giacalone et al. 2021); four-scenario estimates without a bound-companion or diluted-planet scenario give 0.0078 and 0.11, lower bounds. Both remain candidates pending follow-up.

Keywords: Exoplanet detection methods (489), Exoplanets (498), Transit photometry (1709)

1. THE SIGNALS

In the KBonus Kepler Background light curve (Martínez-Palomera et al. 2023a) of KIC 6699887 (Gaia DR3 2077554781499350144, $G = 16.2$), recorded by Kepler (Borucki et al. 2010) only in other targets' pixels, SkySift's box search over 0.5 to 200 day periods found two independent signals. The period ratio is 3.155, 5 percent from 3, outside the 2 percent alias tolerance of the search (ratios of 2, 3, 1/2 and 1/3), so neither is an alias of the other. One of the 31 predicted 47.69 d transits falls within a duration of a 15.11 d transit.

A dip of 1,160 ± 50 ppm lasting $4.02^{+0.13}_{-0.11}$ hours repeats every 15.113696 ± 0.000054 days (SDE 66.0, S/N 32.9, 81 transits), with the reference mid-transit at $\mathrm{BJD_{TDB}}$ 2455662.9571 ± 0.0015 at epoch 46, the centre of the series; 57 of 60 timed transits fit this linear ephemeris with an rms of 18.0 minutes.

A second dip of 993 ± 87 ppm lasting $5.47^{+0.10}_{-0.14}$ hours repeats every 47.69057 ± 0.00046 days (SDE 33.2, S/N 19.2, 28 transits), with the reference mid-transit at $\mathrm{BJD_{TDB}}$ 2455731.7581 ± 0.0041 at epoch 16; 19 of 20 timed transits fit this linear ephemeris with an rms of 24.2 minutes.

2. VETTING

SkySift vets each signal through a fixed ladder of checks; the shape fit is a quadratic limb-darkened model (Claret & Bloemen 2011) on a circular orbit, integrated over the 29.4-minute exposure.

15.11 d: 80 of 96 predicted transits with data are well covered; the fit gives a radius ratio of $0.0357^{+0.0019}_{-0.0014}$, b is 0.80 and a/R* 19.0 fitted, 13.0 sigma from the host's density (no error; ingress shorter than the cadence: b and a/R* are degenerate with k); the odd and even depths are 1,103 ± 64 and 1,209 ± 75 ppm (1.1 sigma apart); the phase 0.5 depth is -12 ± 64 ppm (3-sigma limit 192 ppm); the depth agrees across quarter groups (p = 0.22).

47.69 d: 27 of 31 predicted transits with data are well covered; the fit gives a radius ratio of $0.02856^{+0.00088}_{-0.00092}$, b and a/R* are unconstrained at this cadence (the quoted values follow the host's density): b is below 0.16 (84th percentile) and a/R* $68.9^{+1.1}_{-1.7}$; the odd and even depths are 860 ± 73 and 1,114 ± 154 ppm (1.5 sigma apart); the phase 0.5 depth is 89 ± 77 ppm (3-sigma limit 319 ppm); the depth agrees across quarter groups (p = 0.03).

3. WHICH STAR IS DIMMING

The centroid errors carry a pointing-jitter floor of 0.017 px. The depths are KBonus PSF photometry, decontaminated to first order, with no crowding correction applied, with a CROWDSAP of 0.88 and a PSF flux fraction inside the aperture of 0.73.

For the 15.11 d candidate, the difference-image centroid sits a median 0.16 pixels from this star and 2.47 from its brightest neighbour, nearer this star in 4 of 4 observing seasons (in seasons 1, 2 and 3 this star's 3 x 3 box is cut by

the stamp edge, adding up to 0.18 px to the centroid error); against the star's out-of-transit image the offset is 0.54 ± 0.17 arcsec (3.2 sigma), an offset by the rule of Bryson et al. (2013) (over 3 sigma and 0.2 arcsec); a two-source fit gives a median depth of 1,213 ppm on this star and 23 on the neighbour.

For the 47.69 d candidate, the centroid sits a median 0.20 pixels from this star and 2.50 from its brightest neighbour, nearer this star in 4 of 4 observing seasons (the same boxes are cut); the offset is 0.57 ± 0.43 arcsec (1.3 sigma), not an offset by that rule; the two-source fit gives 1,067 ppm on this star and -12 on the neighbour.

4. THE STAR AND PLANETS

No Kepler stellar catalogue covers a background star, so the parameters come from Gaia DR3 (Gaia Collaboration et al. 2016, 2023) colour and parallax; Gaia gives a RUWE of 1.09 and no non-single-star solution. The reddening is E(B-V) = 0.044 (0.041 to 0.065) from Bayestar19 (Green et al. 2019) at 841 pc (full column 0.113 (Schlafly & Finkbeiner 2011), an upper bound). The colour gives a spectral type of K4V and an effective temperature of $4{,}693^{+192}_{-181}$ K, and the zero-point-corrected parallax a distance of 841^{+26}_{-24} pc. The star has a radius of $0.730^{+0.031}_{-0.024}$ $R_\odot$ and a mass of $0.750^{+0.038}_{-0.037}$ $M_\odot$.

The 15.11 d candidate's radius is $2.85^{+0.19}_{-0.15}$ $R_\oplus$ (ratio 0.0357), on an orbit of 0.1087 ± 0.0018 AU at an equilibrium temperature of 536^{+29}_{-25} K and $19.7^{+4.6}_{-3.4}$ times Earth's insolation.

The 47.69 d candidate's radius is $2.27^{+0.12}_{-0.10}$ $R_\oplus$ (ratio 0.0286), on an orbit of 0.2339 ± 0.0039 AU at an equilibrium temperature of 366^{+20}_{-17} K and $4.26^{+1.00}_{-0.74}$ times Earth's insolation.

5. HOW LIKELY IS A FALSE POSITIVE

6 of 1,998 null searches of other KBonus stars reached the 15.11 d candidate's SDE and 16 of 1,998 the 47.69 d candidate's (1 in 290 and 1 in 120); across the 1,579-star sweep about 6 and 13 such peaks are expected by chance, and the vetting ladder, not the SDE, separates the candidates from them. TRICERATOPS (Giacalone et al. 2021), run on the KBonus light curve with Kepler limb darkening and Gaia DR3 field stars, gives median false-positive probabilities of 0.028 and 0.037 (0.012 and 0.015 without a planet transiting an unresolved bound companion, STP, absent from SkySift's model; 0.019 and 0.024, and 0.0035 and 0.0029 without STP, with the 8 neighbours the difference images exclude removed) and nearby false-positive probabilities of 0.0081 and 0.012 over 20 runs (no validation class applies at 30-minute cadence); the pixel-excluded values rest on our own difference-image centroids, the all-neighbours values on a crowding bound. A four-scenario estimate without a bound-companion or diluted-planet scenario gives 0.0078 and 0.11, lower bounds. TESS observed the star in 10 sectors; the predicted folded significances of 0.46 to 0.93 and 0.25 to 0.54 sigma mean no detection is expected at the nominal noise; no TESS data were examined (Christ et al. 2019).

6. WHAT IS NOT YET KNOWN

A grazing or diluted eclipsing binary is not fully excluded by photometry alone. No spectrum: the planet radius depends on the stellar radius from Gaia colour and parallax. The two candidates were vetted singly; no joint fit was made.

7. DATA AVAILABILITY

Pack (data and methods): doi:10.5281/zenodo.23843993; light curves: the KBonus Kepler Background high-level science product at MAST (Martínez-Palomera et al. 2023b); code: SkySift at commit f542d65, doi:10.5281/zenodo.23843666.

Software: SkySift 0.1.0 (search version 5+6cfe4cf4, commit f542d65), numpy 2.5.4, scipy 1.18.1, astropy 8.0.1, matplotlib 3.11.2, triceratops 1.1.0

Facilities: Kepler, Gaia, MAST, Exoplanet Archive

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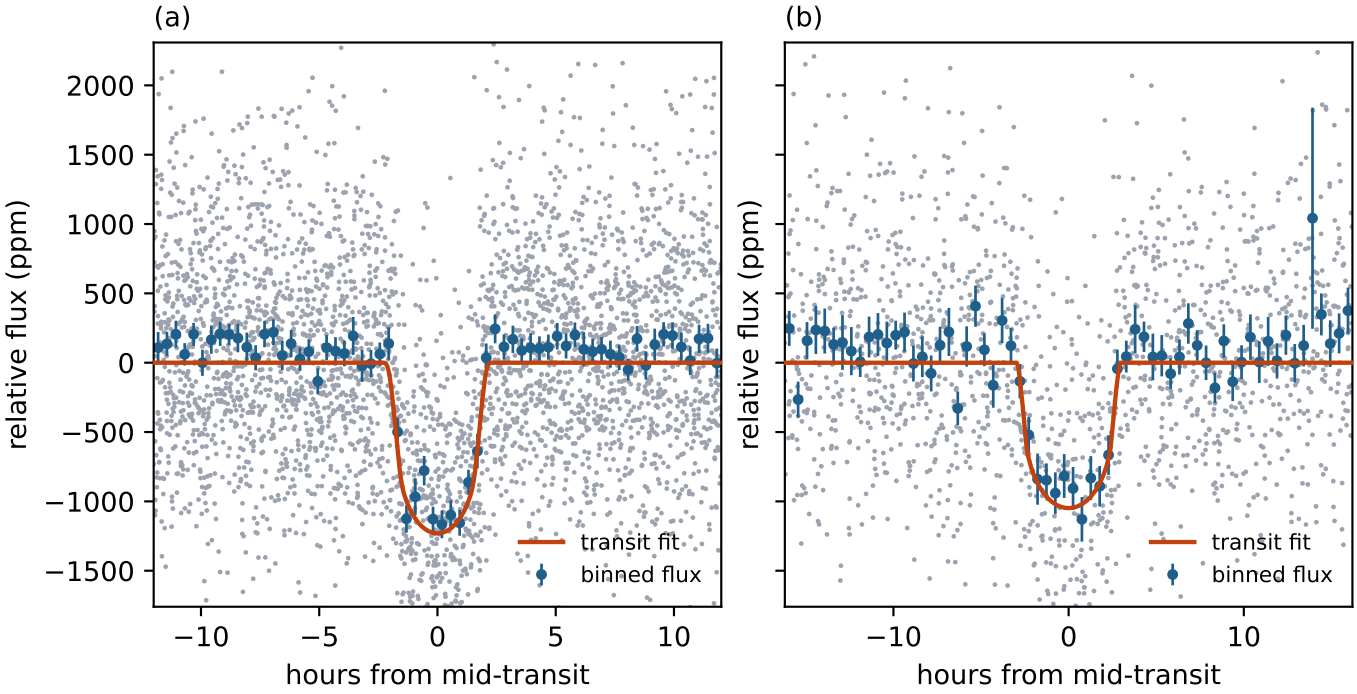


Figure 1. (a) the light curve folded on the 15.113696 day linear-fit ephemeris and (b) on the 47.69057 day one, within 12.0 and 16.2 hours of mid-transit, in parts per million: bins of 22.5 and 30.4 minutes with standard errors (dots) and the transit fits (lines).

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